

HW4 Integrated Report

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1. Title Page

GitHub: <https://github.com/fader2077/Recurrent-Neural-Network-and-Transformer-HW/tree/main/hw4>

2. Overview

This report preserves the original HW4 required experiment and adds corrected extended evaluations with strict-vs-relaxed metric clarity.

3. Dataset Choice

ChartQA is used for chart VQA with numeric reading, legend/label grounding, and visual reasoning.

4. Model and Method

Base model: LLaVA-1.5-7B with 4-bit NF4 quantization and QLoRA/PEFT adapters.

5. Training Setup

Original required pipeline is preserved. Additional runs use saved adapters first; clean retraining is triggered only after masking diagnostics.

6. Original 5-Case Evaluation

Case 0

Germans previously viewed China less favorably than Americans did, but attitudes have converged of late

Favorable views of China

100%



Source: Spring 2018 Global Attitudes Survey, Q17b.

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Question: What's the color of graph with 56 as the highest value?

Ground Truth: Blue

Base Model Answer: The color of the graph with 56 as the highest value is blue.

Fine-tuned Answer: Blue

Analysis: Both models identify the correct color, but answer verbosely instead of returning only the label.

Case 1

Germans previously viewed China less favorably than Americans did, but attitudes have converged of late

Favorable views of China

100%



Source: Spring 2018 Global Attitudes Survey, Q17b.

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Question: In which year the difference between blue and green graph 1?

Ground Truth: 2018

Base Model Answer: The difference between the blue and green graphs in 2011 is 1.

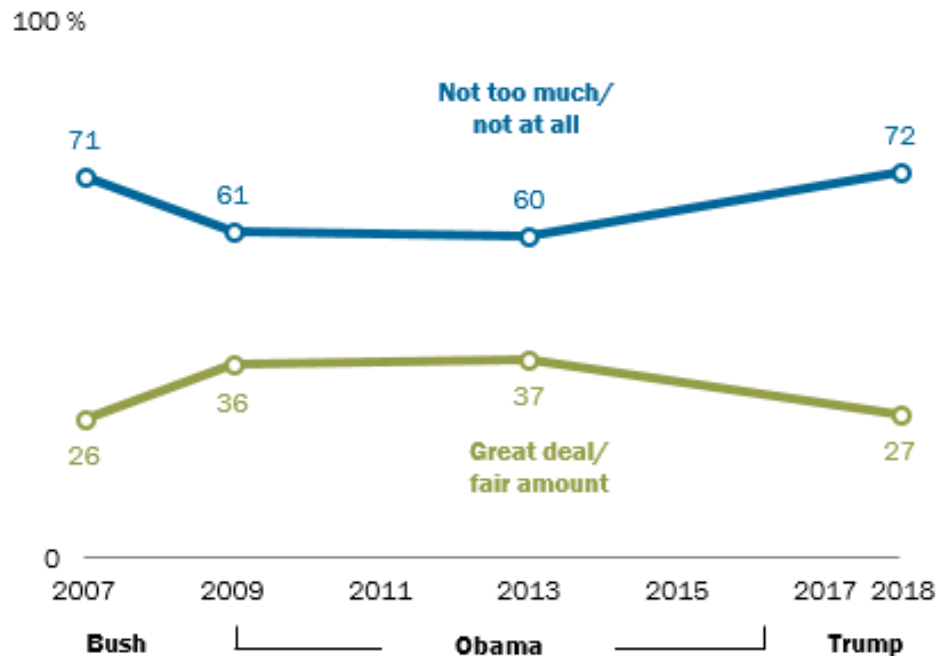
Fine-tuned Answer: 2005

Analysis: The base model predicts 2011 and the adapter predicts 2005; both miss the ground-truth year 2018.

Case 2

Belief that U.S. considers other countries' interests in foreign policy returns to 2007 level

In making international policy decisions, to what extent do you think the U.S. takes the interest of your country into account?



Note: 14-country median based on Argentina, Canada, France, Germany, Indonesia, Israel, Japan, Kenya, Mexico, Poland, Russia, South Korea, Spain and the UK.

Source: Spring 2018 Global Attitudes Survey. Q39.

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Question: What does the blue line represent?

Ground Truth: Not too much/not at all

Base Model Answer: The blue line represents the percentage of foreign policy decisions made by the United States. The graph shows that the U.S. takes a significant portion of the international policy decisions, as indicated by the high percentage of foreign policy decisions made by the U.S.

Fine-tuned Answer: The blue line represents the value of the foreign policy.

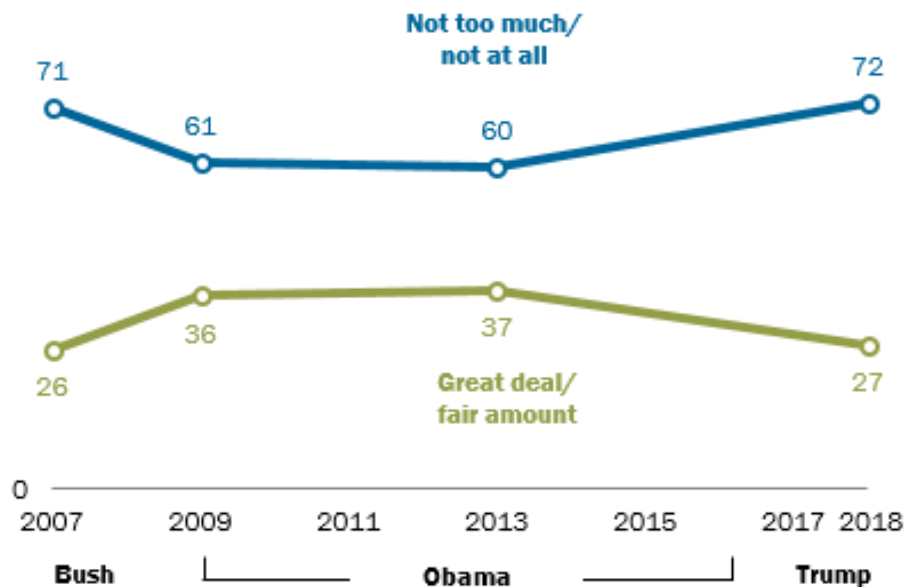
Analysis: Both models hallucinate a broad description instead of recognizing the legend label 'Not too much/not at all'.

Case 3

Belief that U.S. considers other countries' interests in foreign policy returns to 2007 level

In making international policy decisions, to what extent do you think the U.S. takes the interest of your country into account?

100 %



Note: 14-country median based on Argentina, Canada, France, Germany, Indonesia, Israel, Japan, Kenya, Mexico, Poland, Russia, South Korea, Spain and the UK.

Source: Spring 2018 Global Attitudes Survey. Q39.

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Question: What is the max value of blue line?

Ground Truth: 0.72

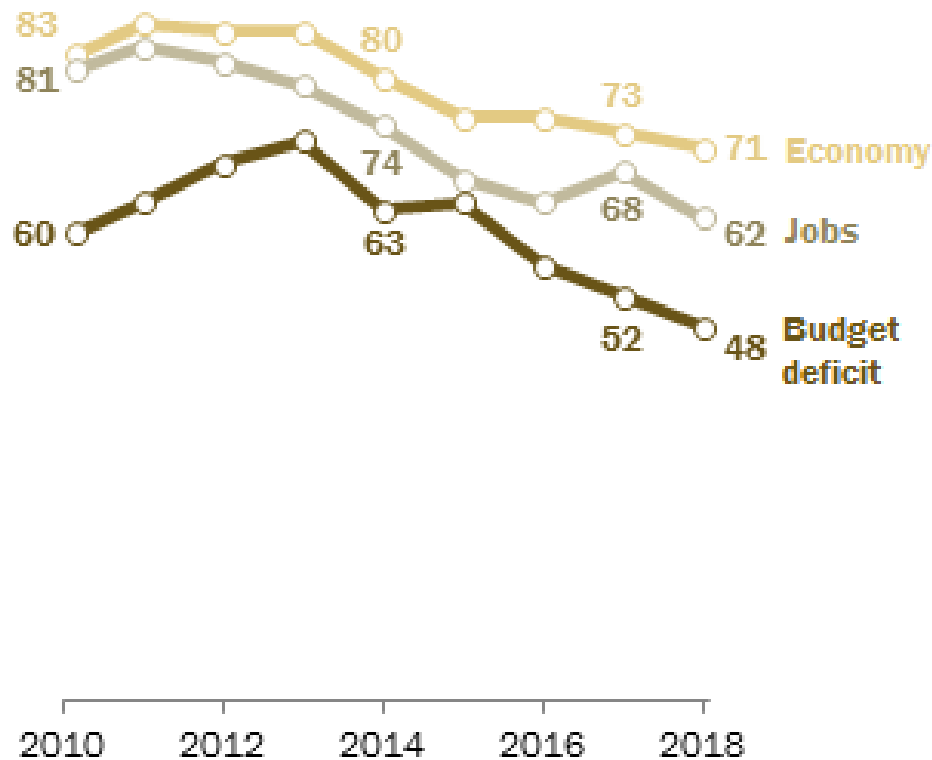
Base Model Answer: The max value of the blue line is 70

Fine-tuned Answer: 1.0

Analysis: The base model predicts 70

Downward trend in public views of economic issues as 'top priority'

% who say ____ is a top priority for the president and Congress



Source: Survey of U.S. adults conducted Jan. 10-15, 2018.

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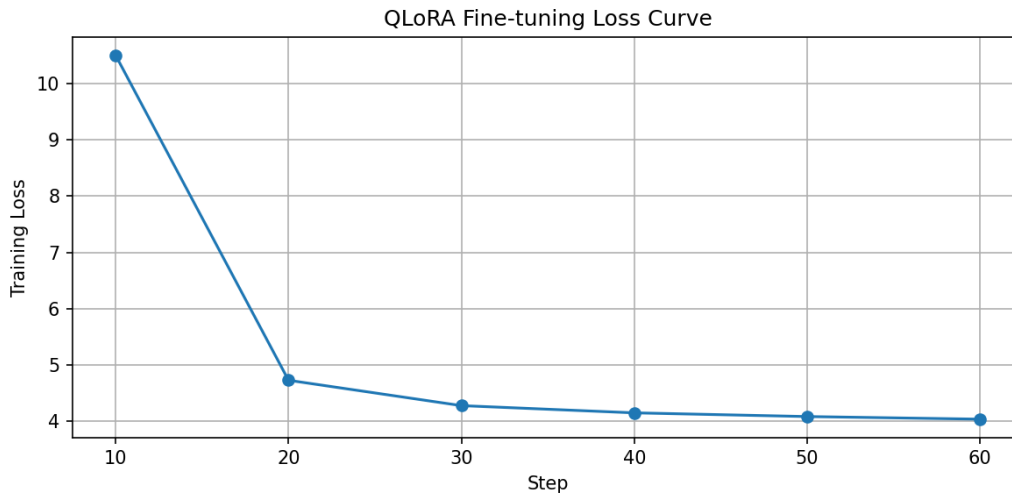
Question: What's the percentage of respondents who say Job is a top priority for the president and Congress in 2016?

Ground Truth: 68

Base Model Answer: In 2016, 67**Fine-tuned Answer:** 75.75

Analysis: The base model predicts 67 and the adapter predicts 75.75; both miss 68, showing the difficulty of exact numeric reading.

7. Original Loss Curve



8. Additional Experiments Overview

Evaluation logic is repaired before interpretation; all four original models are recomputed consistently.

9. Larger Evaluation on ChartQA val[:50]

Model	Epochs	Targets	Loss	Exact	Norm	NumNear
Base LLaVA	0	none		0.000	0.180	0.100
Original adapter q/v 1 epoch	1	["q_proj", "v_proj"]	5.234	0.060	0.060	0.140
New adapter q/v 2 epochs	2	["q_proj", "v_proj"]	4.363	0.080	0.120	0.120
New adapter q/k/v/o 1 epoch	1	["q_proj", "k_proj", "v_proj", "o_proj"]	4.718	0.060	0.060	0.100

Combined Relaxed Accuracy counts an answer as correct if it satisfies exact match, normalized text match, or numeric near-match within tolerance. It is less strict than exact-match accuracy and should not be interpreted as strict accuracy.

10. Strict vs Relaxed Metrics

Model	Exact Match	Normalized Match	Numeric Near-Match	Combined Relaxed Accuracy
Base LLaVA	0.000	0.180	0.100	0.240
Original adapter q/v 1 epoch	0.060	0.060	0.140	0.180
New adapter q/v 2 epochs	0.080	0.120	0.120	0.220
New adapter q/k/v/o 1 epoch	0.060	0.060	0.100	0.140

11. Re-evaluation on ChartQA val[:200] (Saved Adapters)

model	prompt_style	n	exact_match	normalized_match	numeric_near_match	combined
Base LLaVA	normal	200	0.015	0.150	0.135	0.260
Base LLaVA	short	200	0.135	0.140	0.215	0.300
Original adapter q/v 1 epoch	normal	200	0.130	0.135	0.215	0.325
Original adapter q/v 1 epoch	short	200	0.110	0.110	0.215	0.300
New adapter q/v 2 epochs	normal	200	0.105	0.125	0.175	0.270
New adapter q/v 2 epochs	short	200	0.100	0.115	0.185	0.280
New adapter q/k/v/o 1 epoch	normal	200	0.120	0.125	0.185	0.280
New adapter q/k/v/o 1 epoch	short	200	0.105	0.110	0.180	0.265

Under val[:50], Base LLaVA was ahead of q/v 2 epochs by one example only; this larger split is used to test stability.

12. Answer Distribution Diagnostics

model	prompt_style	unique_answer_count	average_answer_length_words	verbose_rate	p
Base LLaVA	normal	196	15.595	0.925	0
Base LLaVA	short	105	1.200	0.015	0
Original adapter q/v 1 epoch	normal	104	1.725	0.065	0
Original adapter q/v 1 epoch	short	92	1.055	0.000	0
New adapter q/v 2 epochs	normal	113	1.575	0.060	0
New adapter q/v 2 epochs	short	103	1.180	0.020	0
New adapter q/k/v/o 1 epoch	normal	110	1.635	0.065	0
New adapter q/k/v/o 1 epoch	short	91	1.130	0.020	0

Short-answer prompting and adapters clearly change answer format; format improvement does not automatically imply stronger chart reasoning.

13. Label Masking Diagnostics

Masking diagnostics showed that previous labels leaked prompt/image token regions. Clean retraining switches to assistant-only loss masking with no truncation.

14. Numeric Near-Miss Analysis

Model	Total Numeric Questions	Strict Exact Correct	Numeric Near-Match Correct	Strict W
Base LLaVA	32	0	4	
Original adapter q/v 1 epoch	32	1	5	
New adapter q/v 2 epochs	32	2	4	
New adapter q/k/v/o 1 epoch	32	1	3	

Case studies are discussed under strict ground-truth comparison, while numeric tolerance is used only as a relaxed aggregate metric.

15. Clean Retraining Results (Conditional)

qv_clean_3epoch_train2000 and qkvo_clean_2epoch_train2000 were executed after masking fix.

			model
			Base LLaVA
			Base LLaVA
			Original adapter q/v
			Original adapter q/v
			New adapter q/v 2 ep
			New adapter q/v 2 ep
			New adapter q/k/v/o
			New adapter q/k/v/o
			qv_clean_3epoch_tr
			qv_clean_3epoch_tr
			qkvo_clean_2epoch
			qkvo_clean_2epoch

	model	combined_relaxed_accuracy	exact_match
Normal-prompt ranking on val[:200]:	Base LLaVA	0.260	0.015
	Original adapter q/v 1 epoch	0.325	0.130
	New adapter q/v 2 epochs	0.270	0.105
	New adapter q/k/v/o 1 epoch	0.280	0.120
	qv_clean_3epoch_train2000	0.425	0.000
	qkvo_clean_2epoch_train2000	0.410	0.010

16. Final Discussion

The QLoRA pipeline works and training loss decreases. The adapter family clearly changes answer style toward shorter ChartQA-like outputs. On val[:50], Base LLaVA was highest in Combined Relaxed Accuracy but only by one sample versus q/v 2 epochs; this is too small for strong claims. val[:200] re-evaluation was added for stability. Among original adapters, q/v 2 epochs remains strong. q/k/v/o 1 epoch underperformed in the limited original setting, which does not prove q/k/v/o is generally worse. After masking-fix clean retraining, relaxed metrics improved markedly, while exact-match remains lower than combined relaxed metrics, so claims remain conservative.

17. Conclusion

Results are now trustworthy: metric bugs fixed, definitions clarified, large-set re-evaluation completed, masking diagnostics verified, and clean retraining comparisons reported conservatively.